

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	A custom Python script was developed to automate data acquisition from open sources, which is publicly available on GitHub at: https://github.com/Yigang-Chen/SCOPE-DTI . The source code has also been deposited to Zenodo at https://doi.org/10.5281/zenodo.17217391 .
Data analysis	The model and all related code for data analysis are publicly available on GitHub: https://github.com/Yigang-Chen/SCOPE-DTI . The source code has also been deposited to Zenodo at https://doi.org/10.5281/zenodo.17217391 .

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The Python scripts and trained models developed in this study are openly available on GitHub at <https://github.com/Yigang-Chen/SCOPE-DTI>. The benchmark datasets used for validation are publicly available as cited in the manuscript. The SCOPE dataset generated in this study has been deposited in Zenodo under accession

code <https://doi.org/10.5281/zenodo.1721746965> and is also accessible via our webserver (<https://awi.cuhk.edu.cn/SCOPE/>).

All other datasets used in this study are publicly available from established resources, including DrugBank (<https://go.drugbank.com>), ChEMBL (<https://www.ebi.ac.uk/chembl>), BindingDB (<https://www.bindingdb.org>), DrugCentral (<http://drugcentral.org>), the Therapeutic Target Database (<http://db.idrblab.net/ttd/>), PubChem (<https://pubchem.ncbi.nlm.nih.gov>), Pharos (<https://pharos.nih.gov>), PROMISCUOUS (<http://bioinformatics.charite.de/promiscuous>), and the IUPHAR/BPS Guide to PHARMACOLOGY (<https://www.guidetopharmacology.org>). Additional resources include TransformerCPI (<https://github.com/lifanchen-simm/transformerCPI>), the SNAP library (<http://snap.stanford.edu>), and previously published kinase inhibitor datasets.

Representative protein structures used for molecular visualization were obtained from AlphaFold DB and the Protein Data Bank, including https://alphafold.ebi.ac.uk/files/AF-P46095-F1-model_v4.pdb, https://alphafold.ebi.ac.uk/files/AF-O43826-F1-model_v4.pdb, https://alphafold.ebi.ac.uk/files/AF-P42224-F1-model_v4.pdb, and <https://files.rcsb.org/download/1MBA.cif>. Additional protein structures were obtained from AlphaFold DB (<https://www.alphafold.ebi.ac.uk/>) and the RCSB PDB (<https://www.rcsb.org/>).

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	None was used in this study.
Reporting on race, ethnicity, or other socially relevant groupings	None was used in this study.
Population characteristics	None was used in this study.
Recruitment	None was used in this study.
Ethics oversight	None was used in this study.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No statistical methods were used to predetermine sample size. For cell viability assays, six biological replicates per concentration were used, which is a standard and sufficient practice for detecting significant differences in in vitro cytotoxicity studies. For CETSA and Western blot experiments, at least three independent biological replicates were performed to ensure reproducibility.
Data exclusions	No data were excluded from any experiments. All obtained data points were included in the final analysis without applying pre-established exclusion criteria.
Replication	All key experiments, including CCK-8 cell viability assays, CETSA-Western blot, and SPR analysis, were repeated independently at least three times. Consistent results were obtained across biological replicates, confirming the reproducibility of the findings.
Randomization	Experimental samples were grouped based on treatment conditions. As cell-based experiments involved homogenous cell populations under controlled in vitro conditions, formal randomization was not applicable but care was taken to minimize technical variability.
Blinding	Investigators were not blinded to group allocation during data collection or analysis. Given the objective nature of the assays (e.g., absorbance measurements, Western blot signal intensities, SPR response units), blinding was not considered necessary or relevant for this in vitro study.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

The following primary antibodies were used, all diluted 1:1000 for Western blotting:: Anti-AKR1B10 antibody [EPR14421] (Abcam, cat. no. ab192865, clone EPR14421, lot no. 1008975-5); Anti-AMPK alpha 2 antibody [EP20772] (Abcam, cat. no. ab214425, clone EP20772, lot no. 1082338-3); Anti-NF-κB p65 antibody [E379] (Abcam, cat. no. ab32536, clone E379, lot no. 106853-53); Anti-CYP3A5 antibody [EPR4396] (Abcam, cat. no. ab108624, clone EPR4396); Beta Actin Polyclonal antibody (proteintech, cat. no. 20536-1-AP, lot no. 00093906);and GAPDH Monoclonal antibody (proteintech, cat. no. 60004-1-Ig, clone 1E6D9, ot no.10017731). The secondary antibodies used were HRP-labeled Goat Anti-Rabbit IgG(H+L) (Beyotime, cat. no. A0208, lot no. 030623230928) and HRP-labeled Goat Anti-Mouse IgG(H+L) (Beyotime, cat. no. A0216, lot no. 011623230919), both used at 1:1000 dilution.

Validation

The antibodies used in this study were validated for species reactivity and application suitability based on manufacturer information and relevant literature. The Rabbit Recombinant Monoclonal Anti-AKR1B10 antibody [EPR14421] (Abcam, catalog no. ab192865, lot no. 1008975-5) is suitable for Western blot (WB), immunocytochemistry/immunofluorescence (ICC/IF), intracellular flow cytometry (Flow Cyt, Intra), and immunohistochemistry on paraffin-embedded sections (IHC-P). It reacts with human samples and has been cited in 4 publications. The Rabbit Recombinant Monoclonal Anti-AMPK alpha 2 antibody [EP20772] (Abcam, catalog no. ab214425, lot no. 1082338-3) is suitable for immunoprecipitation (IP) and Western blot (WB). It reacts with human samples and has been cited in 3 publications. The Rabbit Monoclonal Anti-NF-κB p65 antibody [E379] (Abcam, catalog no. ab32536, lot no. 106853-53) is suitable for Western blot (WB), immunohistochemistry (IHC), immunocytochemistry/immunofluorescence (ICC/IF), flow cytometry, and immunoprecipitation (IP). It reacts with both human and mouse samples. The antibody's specificity was confirmed using RELA knockout cell line validation. Biophysical quality control (QC) ensures consistency across batches. It is validated on the Leica BOND™ RX automated IHC staining platform and has been cited in over 430 publications. A rabbit recombinant monoclonal antibody [EPR4396] (Abcam, cat. no. ab108624), validated for WB, IHC-P, and ICC/IF. 20536-1-AP targets Beta Actin in WB, IHC, IF/ICC, IP, CoIP, ELISA applications and shows reactivity with human, mouse, rat, canine, monkey samples and has been cited in 4590 studies. 60004-1-Ig targets GAPDH in WB, IHC, IF/ICC, FC (Intra), IP, CoIP, ELISA applications and shows reactivity with human, mouse, rat, pig, zebrafish, yeast, plant samples and has been referenced in 13,679 publications. HRP-labeled Goat Anti-Rabbit IgG(H+L) is secondary antibodies used in Western, ELISA, IHC, and has been cited in 459 publications. HRP-labeled Goat Anti-Mouse IgG (H+L) is secondary antibodies used in Western, ELISA, IHC, which has been cited in 450 publications.

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)

HepG2 is a human hepatocellular carcinoma cell line established from a 15-year-old Caucasian male patient. The cell line was purchased from CellCook (cat. no. CC0101) .

Authentication

The sample is authenticated by the supplier to be a pure, uncontaminated human Hep G2 cell line, authenticated through STR profiling and species determination assays.

Mycoplasma contamination

The cell line used in this study was not tested for mycoplasma contamination.

Commonly misidentified lines (See [ICLAC](#) register)

None of the cell lines used in this study are listed as commonly misidentified or cross-contaminated according to the databases of ICLAC.

Plants

Seed stocks

None was used in this study.

Novel plant genotypes

None was used in this study.

Authentication

None was used in this study.